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Electronic apparatus

Abstract

An electronic apparatus includes the first chassis and the second chassis which are relatively and rotatably connected, a flexible board provided astride between the first chassis and the second chassis, and a graphite sheet stacked on the flexible board. The flexible board and the graphite sheet have a substantially S-shaped extra length absorption portion formed in at least one of the first chassis and the second chassis and having a first folding portion and a second folding portion which are curved in directions opposite to each other, are fixed to each other at a front and rear of the extra length absorption portion, and are fixed to each other at an intermediate portion between the first folding portion and the second folding portion, excluding the first folding portion and the second folding portion, by an adhesive tape.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to an electronic apparatus in which a first chassis and a second chassis are rotatably connected.

Description of the Related Art

(2) In recent years, electronic apparatuses such as PCs and smartphones that have a touch panel type liquid crystal display and do not have a physical keyboard are rapidly spreading. This type of display of the electronic apparatus is desired to be large when in use, but it is desired to be small when not in use. Therefore, an electronic apparatus configured such that the chassis can be folded by using a flexible display such as organic Electro Luminescence (EL) has become practical (see, for example, Japanese Unexamined Patent Application Publication No. 2021-015522).

(3) Moreover, in a case in which the amount of information transmitted and received between the chassis is large or the like, and when the general wiring becomes too thick, it is conceivable to use

a flexible board. Since the wiring between the chassis is extended or contracted due to a difference in peripheral length during a rotation operation, in Japanese Unexamined Patent Application Publication No. 2022-121092, a substantially S-shaped folding portion is provided on the flexible board to absorb a change in path length due to rotation of the chassis.

SUMMARY OF THE INVENTION

(4) Meanwhile, when the S-shaped folding portion is provided on the flexible board, the S shape is slightly changed because the path length is absorbed according to the rotation of the first chassis and the second chassis, but in this case, there is a concern that abnormal noise occurs depending on an arrangement aspect of the flexible board.

(5) The present invention has been made in view of the above problems, and an object of the present invention is to provide an electronic apparatus capable of suppressing occurrence of abnormal noise from a flexible board and components accompanying the flexible board due to rotation of the first chassis and the second chassis.

(6) An electronic apparatus according to an aspect of the present invention, in which a first chassis and a second chassis are relatively and rotatably connected by a hinge, the electronic apparatus includes: a flexible board provided astride between the first chassis and the second chassis; and a sheet provided astride between the first chassis and the second chassis together with the flexible board while being stacked on the flexible board, in which the flexible board and the sheet have a substantially S-shaped extra length absorption portion formed in at least one of the first chassis and the second chassis and having a first folding portion and a second folding portion which are curved in directions opposite to each other, are fixed to each other at a front and rear of the extra length absorption portion, and are fixed to each other at an intermediate portion between the first folding portion and the second folding portion, excluding the first folding portion and the second folding portion.

(7) The above-described aspect of the present invention can suppress occurrence of abnormal noise from a flexible board and components accompanying the flexible board due to rotation of the first chassis and the second chassis.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view illustrating a state in which an electronic apparatus according to an embodiment is closed and in a 0-degree posture.

(2) FIG. 2 is a plan view schematically illustrating a state in which the electronic apparatus is opened and in a 180-degree posture.

(3) FIG. 3 is a plan view schematically illustrating an internal structure of the electronic apparatus.

(4) FIG. 4 is an exploded perspective view of the electronic apparatus.

(5) FIG. 5 is a cross-sectional side view of a hinge device and the periphery thereof in the 180-degree posture.

(6) FIG. 6 is a cross-sectional side view of the hinge device and the periphery thereof in the 0-degree posture.

(7) FIG. 7 is a cross-sectional side view illustrating a state of a stacked body provided in a space and along a hinge main body in the 180-degree posture.

(8) FIG. 8 is a cross-sectional side view illustrating a state of the stacked body provided in the space and along the hinge main body in the 0-degree posture.

DETAILED DESCRIPTION OF THE INVENTION

(9) Hereinafter, an embodiment of an electronic apparatus according to the present invention will be described in detail based on the drawings. It should be noted that the present invention is not limited to the embodiment.

(10) FIG. 1 is a perspective view illustrating a state in which an electronic apparatus **10** according to an embodiment is closed and in a 0-degree posture. FIG. 2 is a plan view schematically illustrating a state in which the electronic apparatus **10** illustrated in FIG. 1 is opened and in a 180-degree posture. FIG. 3 is a plan view schematically illustrating an internal structure of the electronic apparatus **10** illustrated in FIG. 2. In FIG. 3, cover members **18A** and **18B** and a thermal module **30** are removed. FIG. 4 is an exploded perspective view of the electronic apparatus **10**. FIG. 5 is a cross-sectional side view of a hinge device **14** and the periphery thereof in the 180-degree posture. FIG. 6 is a cross-sectional side view of the hinge device **14** and the periphery thereof in the 0-degree posture.

(11) As illustrated in FIGS. 1 to 4, the electronic apparatus **10** includes a first chassis **12A**, a second chassis **12B**, the hinge device **14**, and a display **16**. The display **16** extends between the chassis **12A** and **12B**. In the present embodiment, the electronic apparatus **10** exemplifies a tablet PC or a laptop PC that can be folded like a book. The electronic apparatus **10** may have a configuration in which two chassis are relatively and rotatably connected.

(12) The chassis **12A** and **12B** are disposed adjacent to each other. The first chassis **12A** includes a frame member **17A** and the cover member **18A**. The frame member **17A** is a rectangular frame-shaped member forming standing walls formed on three sides other than a first end portion **12Aa** adjacent to the second chassis **12B**. The cover member **18A** is a plate-shaped member that closes a rear surface opening of the frame member **17A** (see also FIG. 5). Similarly, the second chassis **12B** includes a frame member **17B** forming standing walls on three sides other than a second end portion **12Ba** adjacent to the first chassis **12A**, and the cover member **18B** that closes a rear surface opening of the frame member **17B**. Front surface openings of the frame members **17A** and **17B** are closed with the display **16**.

(13) Each of the members **17A**, **17B**, **18A**, and **18B** is made of, for example, a metal member such as stainless steel, magnesium, or aluminum, or a fiber-reinforced resin plate containing reinforcing fibers such as carbon fiber, and the like. That is, the members **17A**, **17B**, **18A**, and **18B** are heat conductive materials and have appropriate heat transfer properties.

(14) The hinge device **14** relatively and rotatably connects the chassis **12A** and **12B** in an angular posture between the 0-degree posture and the 180-degree posture. The hinge device **14** also functions as a rear cover that hides a gap between the end portions **12Aa** and **12Ba** formed in the 0-degree posture illustrated in FIG. 1.

(15) Hereinafter, as for the electronic apparatus **10**, description is made in which the direction in which the chassis **12A** and **12B** are aligned is called an X direction, the direction along the end portions **12Aa** and **12Ba** orthogonal to the X direction is called a Y direction, and the thickness direction of the chassis **12A** and **12B** is called a Z direction. In the Z direction, the side where the display **16** is provided is a front surface, and a side where the cover members **18A** and **18B** are provided is a rear surface. In addition, only for the first chassis **12A** and components thereof, the direction of the end portion **12Aa** in the X direction is an X1 side, and the opposite side thereof is an X2 side.

(16) Furthermore, as for an angular posture between the chassis **12A** and **12B**, description is made in which a state where the chassis **12A** and **12B** are stacked so as to overlap each other in a surface normal direction is called the 0-degree posture (see FIG. 1), and a state where the chassis **12A** and **12B** are aligned in a direction perpendicular to the surface normal direction (X direction) is called the 180-degree posture (see FIGS. 2 and 3). It is possible to call the posture between 0 and 180 degrees by appropriately carving the angle. For example, a state in which the surface normal directions of the chassis **12A** and **12B** are orthogonal to each other is a 90-degree posture. These angles are for convenience of description, and the actual product may of course have angular positions slightly deviated from the exact angular positions indicated by angle numbers.

(17) As illustrated in FIG. 3, the first chassis **12A** is equipped with a motherboard **20**, a communication module **22**, a solid state drive (SSD) **24**, and a battery device **26**. The thermal

module **30** (see FIG. 4) is provided in the first chassis **12A**.

(18) The thermal module **30** is a heat radiation element that receives the heat from a CPU **20a**, the communication module **22**, and the SSD **24**, which are electrical components generating heat, to disperse the heat over a wide range and radiate the heat, and includes a vapor chamber, a graphite sheet, and one or more heat pipes. As illustrated by a reference sign **60** in FIG. 4, one of the heat pipes is provided along an edge of the thermal module **30** on the X1 side so that the heat of the heating element is transferred to a heat pipe **60**.

(19) The motherboard **20** is mounted with an electronic component, for example, a central processing unit (CPU) **20a** and the like. The CPU **20a** is a processing device that performs computing related to primary control or processing of the electronic apparatus **10**. The CPU **20a** is the largest heating element among the electronic components mounted in the electronic apparatus **10**. The communication module **22** performs information processing on wireless communication transmitted and received, for example, via an antenna mounted on the second chassis **12B**. The communication module **22** corresponds to, for example, a wireless WAN or a 5th generation mobile communication system. The SSD **24** is a storage device that uses a semiconductor memory. The first chassis **12A** is mounted with various electronic components other than the motherboard **20**. The communication module **22** and the SSD **24** are heating elements having the heat generation amount only next to the CPU **20a**. The battery device **26** is a secondary battery that serves as a sub-power source for the electronic apparatus **10**.

(20) The second chassis **12B** is mounted with a battery device **32**, a display board **34**, and a sub card **36**. The battery device **32** is a secondary battery that serves as the main power source of the electronic apparatus **10**, is larger than the battery device **26**, and occupies most of the second chassis **12B**. The display board **34** is a control board of the display **16**. The sub card **36** is a board on which, for example, a power button, an external connector conforming to a universal serial bus (USB) standard, and the like are mounted. The second chassis **12B** is mounted with various electronic components other than the battery device **32** and the like.

(21) The battery device **32**, the display board **34**, and the sub card **36** are connected to the motherboard **20** using flexible boards **38**, **40**, and **42** while straddling the end portions **12Aa** and **12Ba**, respectively. Hereinafter, the flexible boards **38**, **40**, and **42** will be typically referred to as the flexible boards **38** unless otherwise particularly distinguished.

(22) The amount of heat generated by the battery device **32**, the display board **34**, and the sub card **36** is smaller than that of the CPU **20a** or the like. Therefore, in the electronic apparatus **10**, the amount of heat generated in the first chassis **12A** is larger than the amount of heat generated in the second chassis **12B**. Therefore, the electronic apparatus **10** has a configuration to equalize heat of each of the chassis **12A** and **12B** by promoting heat transfer between the left and right chassis **12A** and **12B**, and includes heat conductive members **78A** and **78B** (see FIG. 4) and graphite sheets **86**, **88**, and **90** (see FIG. 4). The graphite sheets **86**, **88**, and **90** are sequentially stacked and bonded to a part of the flexible boards **38**, **40**, and **42**. These heat conductive members **78A** and **78B** and graphite sheets **86**, **88**, and **90** will be described below.

(23) In the 0-degree posture illustrated in FIGS. 1 and 6, the chassis **12A** and **12B** are folded in two. The display **16** is a paper-like flexible display formed of an organic EL. In the 0-degree posture, the display **16** is disposed such that a region **R1** on the first chassis **12A** side faces a region **R2** on the second chassis **12B** side illustrated in FIG. 2, and a bending region **R3**, which is a boundary region between the regions **R1** and **R2**, is bent in an arc shape. In the 180-degree posture illustrated in FIGS. 2 and 5, the chassis **12A** and **12B** are disposed side by side with each other. In this case, the display **16** has the regions **R1** and **R2** and the bending region **R3** disposed side by side on an XY plane, and forms a flat plate shape as a whole.

(24) In the display **16**, the region **R1** is relatively fixed to the first chassis **12A**, and the region **R2** is relatively fixed to the second chassis **12B**. Specifically, as illustrated in FIG. 5, a rear surface **16a** of the region **R1** is fixed to the first chassis **12A** via a first plate **44A**, and a rear surface **16a** of the

region R2 is fixed to the second chassis 12B via a second plate 44B.

(25) As illustrated in FIG. 5, the plates 44A and 44B are disposed on the left and right so as to interpose the hinge device 14 therebetween, and support the display 16 with the respective front surfaces 44Aa and 44Ba. The rear surface 16a of the display 16 has the region R1 adhesively fixed to the front surface 44Aa of the first plate 44A and the region R2 adhesively fixed to the front surface 44Ba of the second plate 44B. The plates 44A and 44B have a configuration in which a metal frame made of a magnesium alloy surrounds a carbon fiber reinforced resin plate in which carbon fibers are impregnated with a matrix resin such as epoxy resin and an outer periphery of the rear surface of the carbon fiber reinforced resin plate.

(26) The bending region R3 of the display 16 is relatively movable with respect to the chassis 12A and 12B. In the 180-degree posture, the rear surface 16a of the bending region R3 is supported by the hinge device 14 (see FIG. 5). In the 0-degree posture, the bending region R3 is bent in an arc shape, a part of the rear surface 16a is supported by the hinge device 14, and most of the rear surface 16a is separated from the hinge device 14 (see FIG. 6).

(27) As illustrated in FIGS. 5 and 6, the hinge device 14 of the present embodiment has a hinge main body (hinge) 46, a first support plate 48A, and a second support plate 48B.

(28) The hinge main body 46 is provided at a position straddling the end portions 12Aa and 12Ba of the chassis 12A and 12B, and extends over the substantially entire length in the Y direction along the end portions 12Aa and 12Ba. The cross-section of the hinge main body 46 is symmetrical between the first chassis 12A and the second chassis 12B, and has a trapezoidal shape due to the two inclined surfaces 46a and the top surface 46b between the two inclined surfaces 46a. The hinge main body 46 is a block-shaped component formed of a metal material such as aluminum. The hinge main body 46 supports two hinge shafts aligned in the X direction in the 180-degree posture.

(29) As illustrated in FIGS. 1 and 5, a rear cover component 49 is attached to an outer surface of the hinge main body 46. The rear cover component 49 is a substantially U-shaped plate that matches an outer surface shape of the hinge main body 46. The rear cover component 49 is formed of a heat conductive material, for example, an aluminum alloy, stainless steel, or the like. The rear cover component 49 is a decorative cover for improving the outer surface quality. The flexible boards 38, 40, and 42 pass between the hinge main body 46 and the rear cover component 49 at the position straddling the end portions 12Aa and 12Ba.

(30) In the 180-degree posture illustrated in FIG. 5, the hinge main body 46 is accommodated in the chassis 12A and 12B, and straddles the end portions 12Aa and 12Ba close to each other in the X direction. In the 0-degree posture illustrated in FIG. 6, the hinge main body 46 is disposed so as to close a gap formed between the end portions 12Aa and 12Ba which are largely separated from each other. In this case, deterioration of an appearance design of the folded electronic apparatus 10 is prevented by disposing the rear cover component 49 to the outermost surface (see FIG. 1).

(31) That is, the rear cover component 49 covers the gap formed between the chassis 12A and 12B in an angular posture other than the 180-degree posture (for example, the 0-degree posture or the 90-degree posture) (see FIGS. 1 and 6). As a result, the rear cover component 49 is prevented from exposing the internal components of the chassis 12A and 12B from the gap to an appearance. The rear cover component 49 is disposed so as to straddle the end portions 12Aa and 12Ba close to each other in the X direction in the 180-degree posture, and is accommodated in the chassis 12A and 12B (see FIG. 5).

(32) Next, the support plates 48A and 48B are plates formed of a metal material such as aluminum, and have bilateral symmetrical shapes. The support plates 48A and 48B are provided on a front side of the chassis 12A and 12B, and extend over the substantially entire length in the Y direction along the end portions 12Aa and 12Ba.

(33) The first support plate 48A is disposed between the first plate 44A and the hinge main body 46. The first support plate 48A has an edge portion on a side of the first plate 44A, which is relatively and rotatably connected to a predetermined bracket via a rotation axis. The first support

plate **48A** has an edge portion on a side of the hinge main body **46**, which is relatively movable with respect to the hinge main body **46**. Since the configuration, the mounting structure, and the like of the second support plate **48B** are bilaterally symmetrical with those of the first support plate **48A**, detailed description thereof will be omitted.

(34) The support plates **48A** and **48B** swing according to a rotation operation of the chassis **12A** and **12B**. In the 180-degree posture, the support plates **48A** and **48B** support the rear surface **16a** of the bending region **R3** of the display **16** on the front surfaces thereof. In the angular postures other than 180 degrees, the support plates **48A** and **48B** come into contact with the display **16** in a state in which a gap is provided therebetween and the display **16** or with a slight force that does not deform the display **16** (see FIG. 6). The support plates **48A** and **48B** may be configured to support the bending region **R3** of the display **16** even in an angular posture other than 180 degrees and to correct the shape of the bending region **R3**. As described above, in the 180-degree posture, the support plates **48A** and **48B** stably support the bending region **R3** of the display **16** in a plane, and the support plates **48A** and **48B** do not hinder a bending operation of the bending region **R3**.

(35) A connecting edge portion **50A** along the end portion **12Aa** of the frame member **17A** will be described. The connecting edge portion **50A** is substantially symmetrical with and adjacent to the connecting edge portion **50B** in the X direction along the end portion **12Ba** of the frame member **17B** in the 180-degree posture (see FIG. 5). In addition, the connecting edge portion **50A** and the connecting edge portion **50B** are almost symmetrical in the Z direction and face each other in the 0-degree posture (see FIG. 6). Since the connecting edge portion **50A** and the connecting edge portion **50B** have substantially symmetrical shapes, only the connecting edge portion **50A** will be described.

(36) The connecting edge portion **50A** is a part of the frame member **17A** and has appropriate heat transfer properties as described above. The connecting edge portion **50A** includes a rear surface forming portion **50Aa**, a middle stage portion **50Ab**, and a low stage portion **50Ac**. The rear surface forming portion **50Aa** is a part that forms the rear surface of the first chassis **12A** in a region between the end portion **12Aa** of the cover member **18A** and the end portion **12Aa** of the first chassis **12A**. A low step portion **50Ad** is formed on an inner surface of the rear surface forming portion **50Aa** on the X1 side.

(37) The middle stage portion **50Ab** is a part extending to the X2 side from the rear surface forming portion **50Aa**. The cover member **18A** is fixed to the middle stage portion **50Ab** by an adhesive tape **52**. There is a step between the rear surface forming portion **50Aa** and the middle stage portion **50Ab**, and the rear surface forming portion **50Aa** and the cover member **18A** have substantially the same surface as each other.

(38) The low stage portion **50Ac** is a part that slightly shifts from the middle stage portion **50Ab** to the front surface side and extends to the X2 side. A part of the thermal module **30** is fixed to the low stage portion **50Ac**. In FIG. 5, a space **54** is formed on the front surface side of the middle stage portion **50Ab** and on the X1 side of the low stage portion **50Ac**. A part of the flexible board **38** is accommodated in the space **54** so as to draw an inverted S shape. In addition, there is a similar space at symmetrical positions of the second chassis **12B**, and a part of the flexible board **38** is accommodated so as to draw an S shape. Thus, the flexible board **38** can follow the electronic apparatus **10** without difficulty in response to its deformation from the 0-degree posture to the 180-degree posture. The details of this part will be further described below.

(39) As illustrated in FIGS. 4 and 5, the electronic apparatus **10** includes a first heat conductive member **78A** provided in the first chassis **12A** and a second heat conductive member **78B** provided in the second chassis **12B**. The first heat conductive member **78A** has a graphite sheet **80A** and a cushion member **82A**. The second heat conductive member **78B** has a graphite sheet **80B** and a cushion member **82B**. The heat conductive members **78A** and **78B** are disposed in a substantially bilaterally symmetrical manner. The graphite sheets **80A** and **80B** and graphite sheets **73**, **86**, **88**, and **90**, which will be described below, are graphite sheets, which are allotrope of carbon, obtained

by processing graphite into a sheet shape, and have high thermal conductivity. The graphite sheets **86**, **88**, and **90** are thin and flexible sheets having a thickness of, for example, about 10 μm to 1 mm. These graphite sheets may be replaced with sheet-like members formed of a heat conductive material such as a copper sheet or an aluminum sheet.

(40) As illustrated in FIG. 4, the graphite sheets **80A** and **80B** include the sheets **80Aa** and **80Ba** extending in the Y direction and the three sheets **80Ab** and **80Bb** protruding in the X direction so as to be orthogonal to the sheets **80Aa** and **80Ba**. The sheet **80Aa** is disposed along the end portion **12Aa**, and the sheet **80Ba** is disposed along the end portion **12Ba**. The sheet **80Ab** and the sheet **80Bb** protrude in opposite directions.

(41) The cushion members **82A** and **82B** are formed of a material having flexibility and a certain degree of repulsive force, such as rubber and sponge. The cushion members **82A** and **82B** of the present embodiment are sponges. The cushion members **82A** and **82B** are thicker than the graphite sheets **80A** and **80B** in the Z direction, for example, about 1 to 2 mm. The cushion members **82A** and **82B** are disposed at three intersections of the sheets **80Aa** and **80Ba** and the sheets **80Ab** and **80Bb**, respectively. Thus, the heat conductive members **78A** and **78B** have a bank-shaped swelling portion **84** formed at each position where the cushion members **82A** and **82B** are disposed, in which the graphite sheets **80A** and **80B** are raised.

(42) The graphite sheet **80A** is attached to the rear surface forming portion **50Aa** of the connecting edge portion **50A** and to a front surface side of the middle stage portion **50Ab**. The cushion member **82A** and the swelling portion **84** corresponding to the cushion member **82A** are disposed on the step portion **50Ad**. The graphite sheet **80B** and the cushion member **82B** are provided at positions of the connecting edge portion **50B**, which are substantially symmetrical with the graphite sheet **80A** and the cushion member **82A**.

(43) As illustrated in FIG. 5, the cushion members **82A** and **82B** have the graphite sheets **80A** and **80B**, which are pressed against the rear cover component **49** in the 180-degree posture. Therefore, the heat conductive members **78A** and **78B** are in contact with the rear cover component **49** at a part where the swelling portion **84** is present. Thus, the first frame member **17A** is thermally connected to the second frame member **17B** via the graphite sheet **80A**, the rear cover component **49**, and the graphite sheet **80B**, so that heat is transferred between the first chassis **12A** and the second chassis **12B**. The swelling portion **84** abuts a shoulder portion of the rear cover component **49** so as not to deteriorate the front surface of the rear cover component **49** because a sliding distance with respect to the rear cover component **49** is short when the electronic apparatus **10** is deformed from 0 degrees to 180 degrees. As illustrated in FIG. 6, the graphite sheets **80A** and **80B** are separated from the rear cover component **49** in the 0-degree posture so as not to be visually recognized by the user.

(44) Next, the graphite sheets **86**, **88**, and **90** will be described. As described above, the graphite sheets **86**, **88**, and **90** are sequentially stacked in the flexible boards **38**, **40**, and **42** at least a point from the first chassis **12A** to the second chassis **12B**, and are attached by a heat-transferable adhesive tape or the like. Hereinafter, the graphite sheet **86** and the flexible board **38**, which are typically attached to each other, will be described. The graphite sheet **86** has a strip shape and has the same width as the flexible board **38** in the X direction. The graphite sheet **86** and the flexible board **38**, which are attached to each other, are used as the stacked body **100**.

(45) As illustrated in FIG. 5, the graphite sheet **86** and the flexible board **38** are stacked at a point from the first chassis **12A** to the second chassis **12B**, and thus have the same routing route. Therefore, the stacked body **100** is accommodated so as to draw an S-shape and an inverted S-shape in the space **54** of the first chassis **12A** and a corresponding space of the second chassis **12B**. Further, since the graphite sheet **86** and the flexible board **38** are disposed in a path between the rear cover component **49** and the hinge device **14**, the rear surface side is covered with the rear cover component **49** and is not visually recognized. In addition, the front surface side is covered with the display **16**.

(46) The stacked body **100** has a gap between the low stage portion **50Ac** of the connecting edge portion **50A** and the display **16**, which extends in the X2 direction from the space **54** in the first chassis **12A**. The flexible board **38** passing through the gap is connected to the motherboard **20**. On the other hand, the graphite sheet **86** forms a U-shaped folding portion **86a**, and one end portion **86b** reaches a rear surface of the low stage portion **50Ac**, is stacked and interposed between one end portion **86b** and the low stage portion **50Ac** via a heat-transferable rubber **72** by the heat pipe **60**, and is thermally connected. That is, one end portion **86b** of the graphite sheet **86** is thermally connected to the thermal module **30** and the connecting edge portion **50A**. In the embodiment, one end portion **86b** of the graphite sheet **86** is thermally connected by being interposed between the heat pipe **60** and the low stage portion **50Ac**. However, when the thermal connection state thereof is maintained, the stacking order may be changed by, for example, changing the path of the graphite sheet **86** regardless of the stacking order of these three components.

(47) Further, the stacked body **100** is disposed in the second chassis **12B** while being substantially symmetrically with the inside of the first chassis **12A**, and the other end portion **86c** of the graphite sheet **86** is thermally connected to the frame member **17B**, the connecting edge portion **50B**, or the like, which is a part of the frame member **17B**. Therefore, heat of the heating element, such as the CPU **20a**, is transferred from the thermal module **30** to the heat pipe **60**, which is the end portion thereof, and is transferred to one end portion **86b**, which is stacked and interposed by the heat pipe **60** and the low stage portion **50Ac**, so that heat is transferred from the first chassis **12A** to the second chassis **12B** by the graphite sheet **86**.

(48) Further, as described above, heat of the heat pipe **60** is transferred from the low stage portion **50Ac** to the connecting edge portion **50A**, the first heat conductive member **78A**, the rear cover component **49**, the second heat conductive member **78B**, and the connecting edge portion **50B**. Due to these effects, the electronic apparatus **10** can adjust temperature balance between the first chassis **12A** and the second chassis **12B**. In the first chassis **12A**, the temperature rise is moderately suppressed, so that other cooling means such as a fan can be omitted. By omitting mechanical operation elements such as the fan, the cost can be reduced, the noise can be reduced, and the thickness can be reduced.

(49) Hereinafter, the configuration and effect of the stacked body **100**, which is disposed in the first chassis **12A** and along the hinge main body **46**, will be further described. Since the aspect of the inside of the second chassis **12B** of the stacked body **100** is substantially symmetrical with the inside of the first chassis **12A**, the description thereof will be omitted.

(50) FIG. 7 is a cross-sectional side view illustrating a state of the stacked body **100** which is disposed in the space **54** and along the hinge main body **46** in the 180-degree posture. FIG. 8 is a cross-sectional side view illustrating a state of the stacked body **100** which is disposed in the space **54** and along the hinge main body **46** in the 0-degree posture.

(51) In the space **54** in the first chassis **12A**, the stacked body **100** forms a substantially S-shaped extra length absorption portion **104** having a first folding portion **102a** and a second folding portion **102b** which are arranged side by side and curved in opposite directions. The first folding portion **102a**, which serves as the X1 side, and the second folding portion **102b**, which serves as the X2 side, have a so-called hairpin curve shape.

(52) In the stacked body **100**, the path length becomes longer because the inclined surface **46a** is slightly separated from the space **54** in the 0-degree posture (see FIG. 8), as compared to a case where the path length becomes short if necessary because the inclined surface **46a** of the hinge main body **46** is slightly close to the space **54** in the 180-degree posture (see FIG. 7). In response to the deformation from the 0-degree posture to the 180-degree posture of the electronic apparatus **10**, the stacked body **100** is followed by the extra length absorption portion **104** without difficulty. That is, the extra length absorption portion **104** is slightly larger in the 180-degree posture, but slightly smaller in the 0-degree posture.

(53) In the present embodiment, the stacked body **100** has a V-shape at the front and rear of the

edge portion **49a** of the rear cover component **49** in the 180-degree posture, whereas the V-shape disappears in the 0-degree posture. Since the V-shape corresponds to the loosening of the stacked body **100**, the extra length absorption portion **104** has an effect of absorbing the extra length.

(54) In the 180-degree posture, the extra length absorption portion **104** has a slightly large S-shape, and the first folding portion **102a** and the second folding portion **102b** have a shape substantially point-symmetrical with the intermediate portion **106** between the first folding portion **102a** and the second folding portion **102b**. The stacked body **100** is curved clockwise at the first folding portion **102a** toward the second chassis **12B** from the first chassis **12A**, and is curved counterclockwise at the second folding portion **102b**. That is, the intermediate portion **106** includes an inflection point. In this case, the inflection point is not limited to a predetermined point, and includes a linear case. In the 180-degree posture, the extra length absorption portion **104** forms a slightly large S-shape, and the intermediate portion **106** is substantially linear.

(55) In the 0-degree posture, the extra length absorption portion **104** forms a slightly small S-shape, and has slightly different shape from the first folding portion **102a** and the second folding portion **102b**, which are slightly distorted due to the influence of the support plate **48A** or the like. Even when the extra length absorption portion **104** is slightly distorted, a part including the inflection point in the intermediate portion **106** forms a straight line although it is short.

(56) In the flexible board **38** and the graphite sheet **86** of the stacked body **100**, the front stage fixing point **108a** and the rear stage fixing point **108b** are fixed to each other by the adhesive tapes **110a** and **110b**. The rear stage fixing point **108b** is common to each of the extra length absorption portions **104** in the first chassis **12A** and the second chassis **12B**. The front stage fixing point **108a** is a point that is located at a front stage of the extra length absorption portion **104**, and specifically, a point that faces the low stage portion **50Ac** on the X2 side from the space **54**, based on a direction from the first chassis **12A** toward the second chassis **12B**. The rear stage fixing point **108b** is a point that is located at a rear stage of the extra length absorption portion **104**, that is, at a point facing the top surface **46b** of the hinge main body **46**. As described above, since in the stacked body **100**, the flexible board **38** and the graphite sheet **86** are fixed to each other at the front and rear of the extra length absorption portion **104**, positions thereof are not deviated from each other even if the shape of the extra length absorption portion **104** changes.

(57) Further, the flexible board **38** and the graphite sheet **86** are fixed to each other by an adhesive tape **110c** in the intermediate portion **106**. That is, the intermediate portion **106** is a middle stage fixing portion. The adhesive tape **110c** may be provided at least at a point including the inflection point of the first folding portion **102a** and the second folding portion **102b** in the angular posture from the 0-degree posture to the 180-degree posture. As described above, when the flexible board **38** and the graphite sheet **86** are fixed to each other at the front and rear of the extra length absorption portion **104** and at the intermediate portion **106**, the S-shape is maintained in both the flexible board **38** and the graphite sheet **86**. As a result, the flexible board **38** and the graphite sheet **86** are not separated in the extra length absorption portion **104**, and for example, as indicated by a virtual line in FIG. 7, the graphite sheet **86** does not touch a corner portion of the low stage portion **50Ac** or is not fitted between the low stage portion **50Ac** and the first plate **44A**. Therefore, it is possible to prevent contacting and sliding of the graphite sheet **86** and structures thereof by rotation of the first chassis **12A** and the second chassis **12B**, thereby preventing occurrence of abnormal noise. Since the front surface of the graphite sheet **86** is relatively rough, it is conceivable that sliding noise is likely to occur during sliding, but as described above, abnormal noise is prevented from occurring due to fixation of the intermediate portion **106** with the adhesive tape **110c**.

(58) Since the extra length absorption portion **104** is fixed to only the intermediate portion **106** between the first folding portion **102a** and the second folding portion **102b** by the adhesive tape **110c**, excluding the first folding portion **102a** and the second folding portion **102b**, the thickness of the first folding portion **102a** and second folding portion **102b** is suppressed, and slight deviation of the flexible board **38** and the graphite sheet **86** due to a curvature change is allowed, so that the S-

shape is easily deformed into a large and small size, and an extra length of the stacked body **100** is easily absorbed.

(59) The intermediate portion **106** is a point where there is almost no change in curvature when the S-shape is deformed into a large and small size, and is suitable as a point where the flexible board **38** and the graphite sheet **86** are fixed. In addition, since the intermediate portion **106** is a linear point that includes an S-shaped inflection point, is not bent during stationary, and has no change in curvature even during the operation, it is possible to reliably fix the flexible board **38** and the graphite sheet **86** without peeling.

(60) In the present embodiment, as indicated by the virtual line in FIG. 7, the graphite sheet **86** is mainly prevented from contacting and sliding with other structures, but when the flexible board **38** and the graphite sheet **86** are reversely stacked in the stacked body **100**, the flexible board **38** is prevented from contacting the sliding with the other structures. Generally, the flexible board **38** has a smoother front surface than the graphite sheet **86**, but there is a concern that some sliding noise occurs, and abnormal noise is prevented from occurring due to fixation of the intermediate portion **106** with the adhesive tape **110c**.

(61) The space **54** in which the extra length absorption portion **104** is formed is not limited to the point illustrated in FIGS. 7 and 8, and may be other points inside the first chassis **12A**. The extra length absorption portion **104** is provided in both the first chassis **12A** and the second chassis **12B**, but for example, when a display chassis, which includes a main body chassis with a keyboard and a display, is a laptop PC that is configured rotatably by the hinge, the extra length absorption portion **104** may be provided only in the main body chassis. In other words, the flexible board **38** and the graphite sheet **86** may form the extra length absorption portion **104** in at least one of the two chassis according to design conditions, and may be fixed to each other by the S-shaped intermediate portion **106**. When it is less necessary for heat transfer between the first chassis **12A** and the second chassis **12B** and more signal lines, the stacked body **100** may be configured by stacking another flexible board (sheet) on the flexible board **38** instead of the graphite sheet **86**.

(62) Further, in the electronic apparatus **10** according to the present embodiment, the stacked body **100** of the flexible board **38** and the graphite sheet **86** has sliding sheets **112a** and **112b** provided at the front and rear of the extra length absorption portion **104** at a point facing the first folding portion **102a** and the second folding portion **102b**.

(63) The sliding sheet **112a** is attached to the front surface of the graphite sheet **86** on a front stage side, and the sliding of the second folding portion **102b** with the graphite sheet **86** is smoothed to suppress friction noise occurred between the second folding portion **102b** and the graphite sheet **86**. When the graphite sheets **86** are slid against each other, sliding noise is likely to occur, but abnormal noise can be suppressed by interposing the sliding sheet **112a** therebetween.

(64) The sliding sheet **112b** is attached to the front surface of the flexible board **38** on a rear stage side, and the sliding of the first folding portion **102a** with the flexible board **38** is smoothed to suppress friction noise occurred between the first folding portion **102a** and the flexible board **38**. The sliding noise between the flexible boards **38** is smaller than the sliding noise between the graphite sheets **86**, but abnormal noise can be further suppressed by interposing the sliding sheets **112b** between the flexible boards **38**.

(65) Since the sliding sheets **112a** and **112b** are not provided on the folding portions **102a** and **102b** having a change in shape, and are provided on the chassis side having no change in curvature, the sliding sheets **112a** and **112b** are less likely peeled off, so that the change in shape of the folding portions **102a** and **102b** is not hindered.

(66) A sliding sheet **112c** is provided on the graphite sheet **80A** of the first heat conductive member **78A** over the substantially entire length. The sliding sheet **112c** is interposed between the graphite sheet **80A** and the graphite sheet **86** of the stacked body **100** to suppress friction noise occurring therebetween.

(67) The sliding sheets **112a** to **112c** are sheets with excellent sliding properties and wear

resistance, and are, for example, Teflon (registered trademark) (i.e., polytetrafluoroethylene). Although not illustrated, for example, as fixing means of the sliding sheets **112a** to **112c**, an adhesive tape may be used.

(68) Furthermore, in the electronic apparatus **10** according to the present embodiment, the stacked body **100** is fixed to two trapezoidal inclined surfaces **46a** formed by the hinge main body **46** and the top surface **46b** provided between the two trapezoidal inclined surfaces **46a**, by adhesive tapes **114**. Each of the three adhesive tapes **114** is provided at a part except for the trapezoidal corner portion **46c**, and is discontinuous. In FIG. 7, among the adhesive tapes **114**, the reference sign **114a** is added to an adhesive tape that is attached to the inclined surface **46a**, and the reference sign **114b** is added to an adhesive tape that is adhered to the top surface **46b**.

(69) The stacked body **100** has flexibility and some bending rigidity, and has a linear shape in an initial state before assembly to the electronic apparatus **10**, and after assembly, a slight elastic force is generated in a point facing the corner portion **46c** to return to the straight line. In contrast, since the stacked body **100** is fixed to the hinge main body **46** by the adhesive tape **114** so as to be discontinuous at the corner portion **46c**, the thickness and the bending rigidity of the point facing the corner portion **46c** are suppressed, thereby maintaining the suitable flexibility. Therefore, the stacked body **100** is adhesively fixed to the trapezoidal hinge main body **46** without difficulty along one inclined surface **46a** to the top surface **46b** to the other inclined surface **46a**, and is not peeled off.

(70) Further, in the electronic apparatus **10**, the chassis **12A** and **12B** rotate in the angular posture between the 0-degree posture and the 180-degree posture, so that the stacked body **100** has repeated bending stress at a point facing a skirt portion **46d** of the hinge main body **46**. If the stacked body **100** is fixed to a part including the point facing the corner portion **46c** by the adhesive tape **114**, the thickness and rigidity are increased to generate a force to peel off in the part, and the bending stress of the skirt portion **46d** acts synergistically, so that there is a concern that the stacked body **100** is peeled off from the inclined surface **46a** as indicated by the virtual line in FIG. 7. Then, the stacked body **100** is in a state of being fixed to the hinge main body **46** only by the adhesive tape **114b**, and abnormal noise occurs by repeatedly contacting with and separating from the inclined surface **46a** or the rear cover component **49**. In the present embodiment, the stacked body **100** is not peeled off at the point facing the corner portion **46c** as described above, and no abnormal noise occurs.

(71) When the stacked body **100** is not fixed to the hinge main body **46** at the point facing the corner portion **46c**, the adhesive tape **114a** of the inclined surface **46a** is prevented from being peeled off, so that the adhesive tape **114b** of the top surface **46b** may be omitted. However, the stacked body **100** is further reliably fixed by the adhesive tape **114b**, and positioning with respect to the hinge main body **46** becomes easy in assembly.

(72) The present invention is not limited to the above-described embodiment, and there is no doubt that the present invention can be freely changed without departing from the gist of the present invention.

Claims

1. An electronic apparatus in which a first chassis and a second chassis are rotatably connected by a hinge, the electronic apparatus comprising: a flexible board between the first chassis and the second chassis; and a sheet between the first chassis and the second chassis together with the flexible board, wherein the sheet is stacked on the flexible board, wherein the sheet and the flexible board form a stacked body, wherein the flexible board and the sheet have a substantially S-shaped extra length absorption portion in at least one of the first chassis and the second chassis and further have a first folding portion and a second folding portion, both of which are curved in directions opposite to each other, and both of which are fixed to each other at a front and a rear of the extra length absorption portion, and both of which are fixed to each other at an intermediate portion

between the first folding portion and the second folding portion, wherein the stacked body includes a sliding sheet at a point where the stacked body faces the first folding portion at the front and rear of the extra length absorption portion and at a point where the stacked body faces the second folding portion.

2. The electronic apparatus according to claim 1, wherein the sheet is a graphite sheet or a metal sheet.

3. The electronic apparatus according to claim 1, wherein the sliding sheet is polytetrafluoroethylene.

4. The electronic apparatus according to claim 1, wherein the hinge has a trapezoidal cross-section that is symmetrical between the first chassis and the second chassis, and the stacked body is at a point excluding a corner portion of each of two trapezoidal inclined surfaces of the hinge.

5. The electronic apparatus according to claim 4, wherein the stacked body is fixed by an adhesive to a top surface of the hinge and is between the two inclined surfaces of the hinge.
